

Chloe Graham-Major

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MEng Electrical Engineering student at Durham University with experience in commercial analytics, business transformation and quantitative modelling across global energy and infrastructure organisations. Proven ability to translate complex data into actionable insights and support commercial decision-making through rigorous analysis. Strong interest in power market analysis, trading and energy systems, with proficiency in Python, Excel, Power BI, SQL, & MATLAB.

PROFESSIONAL EXPERIENCE

Performance Analytics & Data Enablement Intern (Downstream Commercial)

Shell | *Jun 2026 - Sep 2026*

- Established a data-driven prioritisation framework for global sales transformation initiatives, converting qualitative process-assessment data into an analytical model and Power BI reporting suite used to evaluate 250+ improvement opportunities based on business value, implementation complexity, strategic impact and organisational dependencies.
- Led analysis of 13 global sales processes comprising 182 process steps and 71 pain points, synthesising stakeholder feedback from APAC, EMEA and the Americas to develop 217 improvement opportunities and future-state recommendations spanning AI, automation and commercial process optimisation.
- Presented analytical findings, prioritisation metrics and transformation recommendations to senior stakeholders, challenging existing ways of working and supporting the design of future-state operating models for Shell's CRM transformation programme.
- Built a network of 50+ professionals across Shell Trading & Supply, Shell Energy and Downstream Commercial, gaining insight into power markets, trading, optimisation, risk management and commercial strategy through discussions with traders, analysts and business leaders.

Project Management Consultant Intern

Mace Consult & Network Rail (North-Western & Central Civils & Renewals) | *Jun 2024 - Aug 2024*

- Developed Excel-based reporting and data analysis tools that consolidated project data from multiple sources, improving visibility of programme performance and supporting delivery decisions across live rail infrastructure projects.
- Evaluated project risks, environmental constraints and delivery requirements to support pre-construction planning and contractor mobilisation.
- Applied NEC4 contract principles, commercial governance and project controls, gaining exposure to compensation events and infrastructure programme management.

Civil Engineer | Barnsley Marshall Ltd | *Aug 2025 - Sep 2025*

- Delivered AutoCAD designs and technical drawings and reporting work (including flood risk and drainage strategies) for multiple development projects and bid submissions, including for four Battery Energy Storage System (BESS) sites.
- Prepared fee proposals and lifecycle cost assessments, incorporating commercial considerations into engineering and project feasibility decisions.

EDUCATION

Durham University (Grey College)

MEng Electrical Engineering (Final Year) | *Sept 2023 - July 2027*

Relevant Modules: Electrical Networks and Renewable Integration, Smart Energy Networks and Electrical Energy Conversion, focusing on power system operation, generation economics, electricity market dynamics, network constraints and energy system optimisation. Complemented by studies in Engineering Mathematics, Control & Signal Processing, and Artificial Intelligence & Deep Learning.

Computational Tools for Engineers & Scientists: Optimisation, Monte Carlo simulation, statistical analysis and database analytics using Python, SQL, and MATLAB, including the analysis of NASA MERRA-2 meteorological datasets and large-scale engineering data.

Dissertation Project (2026–27): Temperature-Irradiance-Spectrum Performance Modelling for Perovskite and Organic Photovoltaics

- Developing and validating Python and MATLAB models for emerging photovoltaic technologies, evaluating how uncertainty and modelling assumptions affect energy-yield predictions, Levelized Cost of Energy (LCOE) assessments and techno-economic analysis.
- Performing scenario analysis, sensitivity analysis and uncertainty quantification to evaluate how modelling assumptions influence energy yields.

Electrical Power Systems Lead, Quantum Payload Transmission CubeSat

- Led development of the Electrical Power System for a quantum communications CubeSat, designing battery storage, solar generation and power distribution architecture through energy modelling and power budget analysis, contributing to a First-Class project mark of 70%.
- Designed, assembled and tested custom PCB hardware through simulation, integration and prototype validation, while evaluating sustainability, supply chain and commercial feasibility considerations.

Project Manager, Pedal-Powered Water Purification System

- Led a six-person engineering team in the design and delivery of an off-grid water purification system, achieving a First-Class project mark of 76% through effective project management, stakeholder engagement and technical feasibility analysis.

King Edward VI High School for Girls 2016 - 2023

- **A Levels:** Mathematics (A), Physics (A), Chemistry (A), English Literature (B)

British Science Association CREST Gold and Silver Awards in Engineering | (2023) Piezoelectric Energy Harvesting Systems; (2021) Hydroelectric Energy Recovery from Wastewater

LEADERSHIP & IMPACT

President, Durham University Gospel Choir | 2024 - 25

- Lead the executive committee and oversaw strategic planning, operations and event delivery.
- Managed stakeholder relationships across university departments, external organisations and 40+ society members.

Engineering Social Media Ambassador | 2024 - 26

Expanded the department's social media reach by introducing new content formats and launching its TikTok presence, increasing Instagram reach by 281% and generating 13,000+ views on TikTok.

TECHNICAL SKILLS

Python (Pandas, NumPy, SciPy, scikit-learn, OpenCV), MATLAB, SQL | Excel, Power BI | LTSpice, KiCad, AutoCAD, SolidWorks

Languages: English (Native), French (Intermediate B1), Spanish (Elementary A2), Italian (Elementary A2)